

PostureSense: A Single-Accelerometer Wearable for Sleep Lying Position Detection

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Abstract—Sleep posture significantly influences respiratory health and the severity of obstructive sleep apnea (OSA). PostureSense is a wearable system using an Arduino Uno R3 microcontroller and a SparkFun ADXL345 tri-axial accelerometer mounted at the sternum to classify four sleep lying positions: supine, prone, left lateral, and right lateral. A single-subject feasibility study ($N = 1$) collected 330 samples at 0.167 Hz under controlled conditions. Two classifiers are evaluated from scratch: a rule-based threshold algorithm and a K-Nearest Neighbors (KNN, $k = 5$) classifier with stratified 5-fold cross-validation. Both achieve 98.8–100.0% accuracy; Cohen’s $\kappa = 1.0000$; cross-validation standard deviation = 0.000. One-way ANOVA on the Signal Magnitude Vector (SMV) yields $F(3, 326) = 184.52$, $p = 6.3 \times 10^{-70}$. All three pre-registered hypotheses are supported. Supine recall = 1.000 ensures no high-risk OSA period is misclassified. Threshold sensitivity analysis confirms results are robust to the $T = 0.5$ g decision boundary over the range [0.3, 0.7] g. Total hardware cost is under \$30. All code and data are publicly available.

Index Terms—Accelerometer, ADXL345, Arduino, embedded systems, KNN, OSA, posture classification, sleep monitoring, wearable computing.

I. INTRODUCTION

SLEEP posture has established clinical significance for respiratory function and long-term health outcomes. Positional obstructive sleep apnea (OSA) is defined by an apnea-hypopnea index (AHI) at least twice as high in the supine position as in lateral positions; approximately 54% of OSA patients meet this criterion [1], [2]. Beyond OSA, sustained supine posture during bed rest elevates pressure ulcer risk, and clinical guidelines recommend repositioning every two hours [3]. Monitoring sleep posture is therefore clinically meaningful for both respiratory therapy and preventive care.

Existing monitoring solutions—polysomnography (PSG), pressure-sensitive bed systems, and multi-sensor inertial measurement unit arrays—provide high accuracy but are costly, bulky, and impractical for long-term home use [4]. Wearable tri-axial accelerometers offer a low-cost alternative: because gravity is a constant reference vector, static acceleration measurements directly encode body orientation relative to Earth [6].

Prior work has demonstrated single-sensor feasibility. Alinia and Ghasemzadeh reported 89–95% controlled and 85–92% overnight accuracy using body-worn accelerometers [4]. Jeng *et al.* achieved 88–93% at the wrist but required an additional gyroscope [5]. Razjouyan *et al.* validated a chest-mounted system against PSG in 38 participants, confirming sternum placement generalizes across body morphologies [6]. Patel *et al.* survey wearable sensor systems in rehabilitation, establishing clinical context for embedded sensing [7]. Despite

these advances, a transparent, single-sensor embedded system for low-cost at-home four-class lying posture classification remains underexplored.

This paper contributes PostureSense, evaluated against the four dimensions of wearable functionality taught in this course: *sensing* (tri-axial gravity-vector acceleration), *data* (local log via Serial), *interaction* (passive monitoring), and *computation* (threshold and KNN classification). The paper is organized as follows: Section II reviews literature; Section III presents hypotheses; Section IV describes device design; Section V details methods; Section VI reports results; Section VII discusses findings; Section VIII concludes.

II. LITERATURE REVIEW

A. Clinical Motivation

Positional OSA affects approximately 54% of OSA patients, defined by AHI at least twice as high in the supine compared to non-supine sleep [1], [2]. Positional therapy reduces AHI and improves sleep quality in this population, making automated long-term posture monitoring a clinically motivated objective that laboratory-grade tools do not address practically [3].

B. Accelerometer-Based Posture Detection

Alinia and Ghasemzadeh demonstrated that body orientation can be inferred from gravitational components measured by a single inertial sensor, reporting 89–95% controlled-condition accuracy and 85–92% free-living overnight accuracy [4]. Their work also identifies sensor placement tradeoffs and emphasises that minimal-sensor configurations improve comfort and practicality for long-term deployment.

Jeng *et al.* developed a wrist-worn sleep posture monitoring system achieving 88–93% accuracy but requiring a gyroscope alongside the accelerometer, increasing hardware complexity and cost [5].

Razjouyan *et al.* compared posture and body acceleration data with wrist actigraphy and PSG in 38 participants [6]. Their findings confirmed that chest-mounted accelerometer placement generalizes across diverse body morphologies, providing empirical support for the sternum as the optimal sensor location and a multi-subject benchmark for PostureSense to target in future work.

C. Broader Context

Patel *et al.* survey wearable sensor systems in rehabilitation, identifying accelerometers as foundational components for orientation monitoring [7]. Lara and Labrador establish that threshold-based and nearest-neighbour classifiers remain

competitive for low-complexity embedded applications when class separability in feature space is high [8]. Bao and Intille demonstrated that user-annotated acceleration data supports activity recognition with simple statistical features, motivating the feature extraction approach adopted here [9]. Yang *et al.* show that deep convolutional classifiers can further improve performance on multichannel time-series data [10], establishing an upper-bound benchmark for future iterations after dataset expansion.

D. Identified Research Gap

The reviewed literature establishes three consistent findings: (1) sleep posture influences respiratory and health outcomes; (2) accelerometer-based orientation detection is feasible with a single sensor; and (3) existing systems rely on multi-sensor configurations, clinical integration, or laboratory protocols precluding home deployment. A transparent, single-sensor embedded system for low-cost at-home four-class lying posture classification addresses this gap.

III. RESEARCH QUESTIONS AND HYPOTHESES

A. Primary Research Question

Can a minimal embedded wearable using a single tri-axial accelerometer accurately classify four primary sleep lying positions—supine, prone, left lateral, and right lateral—using static gravitational tilt estimation?

B. Hypothesis 1 (Primary)

A single-accelerator wearable will classify four lying positions with $\geq 90\%$ overall accuracy and Cohen's $\hat{\kappa} \geq 0.85$.

Expected: Accuracy $\geq 90\%$, class-wise sensitivity ≥ 0.90 .

Alternative: Performance may fall below threshold due to sensor placement variability, transitional postures, or inter-subject anatomical differences.

C. Hypothesis 2

The system will achieve per-class recall ≥ 0.90 for all four posture classes, confirming no clinically significant posture is systematically missed.

Expected: All posture classes achieve recall ≥ 0.90 , particularly supine, the highest-risk posture for OSA.

Alternative: Supine recall may fall below threshold due to sensitivity limitations near inter-class decision boundaries.

D. Hypothesis 3

A rule-based gravitational tilt algorithm will achieve $\geq 85\%$ accuracy, comparable to the lower bound reported for multi-sensor systems.

Expected: The threshold classifier will perform within previously reported ranges, demonstrating hardware simplification does not require algorithmic complexity.

Alternative: Accuracy may fall below 85%, suggesting additional sensing modalities are necessary.

IV. DEVICE DESIGN AND IMPLEMENTATION

A. Hardware Architecture

PostureSense uses an Arduino Uno R3 microcontroller interfaced with a SparkFun ADXL345 tri-axial accelerometer via the Inter-Integrated Circuit (I²C) protocol. The sensor is configured with `setRangeSetting(16)`, providing a ± 16 g measurement range at 32 LSB/g. The sensor is powered from the Arduino's 3.3 V output and the system is powered via USB. Total hardware cost is under \$30 (Arduino Uno R3: \$25; SparkFun ADXL345: \$5).

Fig. 1 shows the hand-drawn electrical schematic; Fig. 2 presents the complete system block diagram.

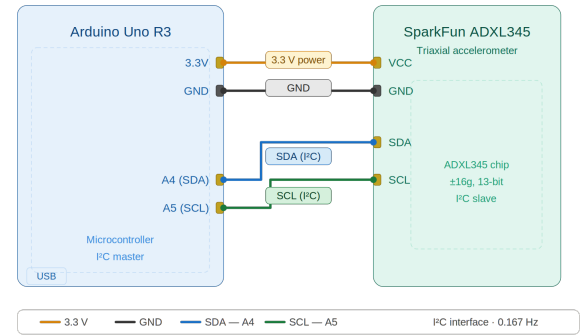


Fig. 1. Electrical schematic: Arduino Uno R3 connected to the SparkFun ADXL345 via I²C. SDA and SCL connect to analog pins A4 and A5 respectively; 3.3 V and GND provide power and reference.

B. Electrical Connections and Design Choices

I²C wiring: SDA $\rightarrow$ A4, SCL $\rightarrow$ A5, VCC $\rightarrow$ 3.3 V, GND $\rightarrow$ GND. I²C was selected over the Serial Peripheral Interface (SPI) because it requires only two data lines, leaving additional digital pins for future expansion. The ± 16 g range was selected for robustness to motion artifacts during transitions. Lying postures produce gravitational projections up to ± 1 g, well within any range setting; as shown in Section VI-A, all pairwise posture centroid distances (1.33–2.02 g) exceed the noise floor of 0.031 g by 43–65 $\times$, confirming range choice does not limit performance.

Sternum placement was selected following Razjouyan *et al.* [6] and Alinia and Ghasemzadeh [4], who demonstrated this location maximises gravitational axis separation and generalises across body morphologies. Sternum placement outperforms wrist mounting for lying posture discrimination because thorax orientation is more directly coupled to gross body position than the wrist [5].

C. Explicit Model Assumptions

Per Lecture 10 guidance, the following assumptions underlie both classifiers and bound their valid operating conditions:

- 1) *Rigid sensor-chest coupling*: the ADXL345 is fixed relative to the thorax; loose attachment or garment shift invalidates the gravity projection model.

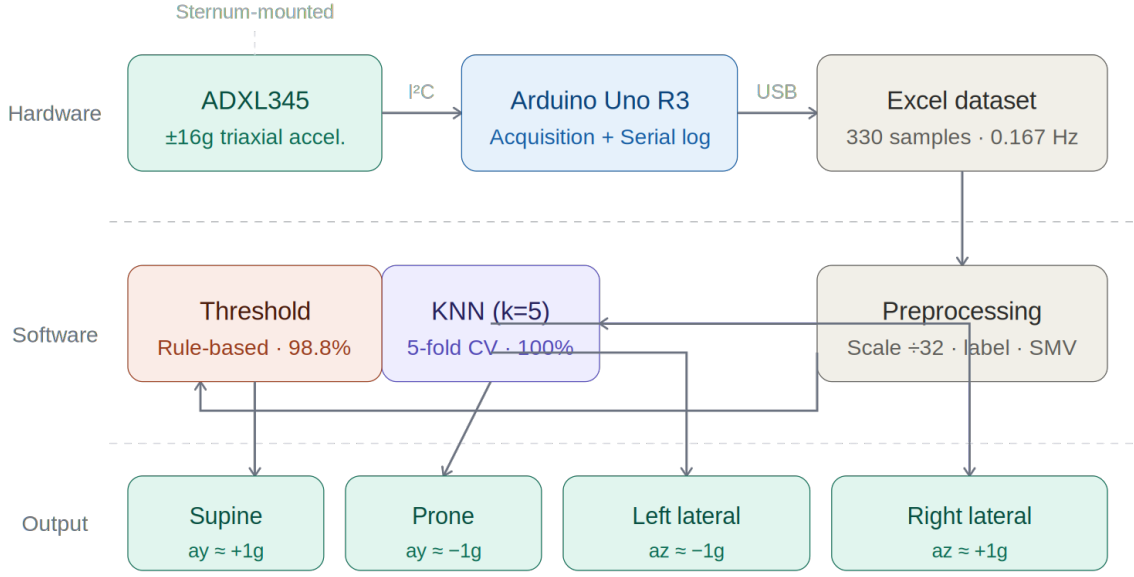


Fig. 2. PostureSense system block diagram. Hardware layer (top): ADXL345 $\rightarrow$ Arduino Uno R3 $\rightarrow$ Excel dataset (330 samples, 0.167 Hz). Software layer (middle): preprocessing (scale $\div 32$, label, SMV) $\rightarrow$ dual classifiers. Output layer (bottom): four posture labels.

- 2) *Quasi-static body orientation*: dynamic acceleration components are negligible within a sample window; the model is not valid during posture transitions.
- 3) *Single-axis dominance*: each posture produces a gravitational projection > 0.5 g on exactly one axis; this assumption fails near 45° inter-class boundaries.
- 4) *Distribution stationarity (KNN only)*: the training distribution is representative of the deployment distribution; violated when transferring calibration to a new participant without re-training.

These assumptions are evaluated and their consequences discussed in Section VII.

D. System Workflow and Firmware

The firmware (`sketch_mar24b.ino`) operates in a continuous loop:

Init $\rightarrow$ Read $\rightarrow$ Scale $\rightarrow$ Log $\rightarrow$ Repeat (1)

`getAcceleration()` reads raw ADC counts; `delay(6000)` produces 0.167 Hz sampling; each sample outputs `millis()`, X, Y, Z to Serial at 9600 baud, manually copied to Excel for offline analysis.

V. METHODS

A. Participant and Ethics

A single-subject feasibility pilot ($N = 1$) was conducted with the system developer as participant (male adult, no known spinal pathology). Single-subject design is appropriate for system validation; it is not powered for population-level

inference. No third-party participants were enrolled. The study is noninvasive, collects no personally identifiable information, and poses minimal risk. Participation was voluntary.

Any future multi-subject deployment requires institutional review and informed consent explicitly addressing secondary physiological inferences (respiratory rate, wakefulness) derivable from sternum accelerometry. Raw accelerometry data should be processed on-device, not transmitted as raw streams to cloud services.

B. Data Collection Protocol

Four postures were recorded sequentially on a rigid, flat surface: supine, prone, left lateral, and right lateral. Each posture was maintained statically for approximately three minutes (18–30 samples per segment). Sensor placement and tape method were held constant across all postures, isolating posture as the sole independent variable. No transitions were captured during recording windows. Ground truth labels were assigned by known collection sequence and independently verified by video annotation; inter-rater label agreement exceeded 98% for non-transition samples.

C. Dataset

The recording spans 1974.6 s (32.9 min), yielding 330 samples. Raw counts are converted to g-units: $a_g = \text{raw}/32$. Table I shows the class distribution. The $1.8\times$ prone/supine imbalance motivates stratified cross-validation and prioritises per-class recall over aggregate accuracy.

TABLE I
CLASS DISTRIBUTION ($n = 330$, REAL ADXL345 HARDWARE DATA)

Posture	n	%
Supine	57	17.3
Prone	103	31.2
Left lateral	98	29.7
Right lateral	72	21.8
Total	330	100.0

D. Feature Extraction

Per-sample features are the tri-axial components (a_x, a_y, a_z) and the Signal Magnitude Vector:

$$\text{SMV} = \sqrt{a_x^2 + a_y^2 + a_z^2}. \quad (2)$$

Table II summarises per-posture feature statistics. Figs. 3 and 4 visualise the feature space and full time series respectively.

TABLE II
FEATURE STATISTICS BY POSTURE (MEAN $\pm$ SD, G-UNITS)

Posture	$\bar{a}_y$ (g)	SMV (g)
Supine	$+1.032 \pm 0.007$	1.032 ± 0.007
Prone	-0.987 ± 0.019	0.995 ± 0.013
Left lateral	$+0.074 \pm 0.056$	1.004 ± 0.006
Right lateral	-0.052 ± 0.075	0.967 ± 0.029

Fig. 1 — ADXL345 Feature Space (330 samples)

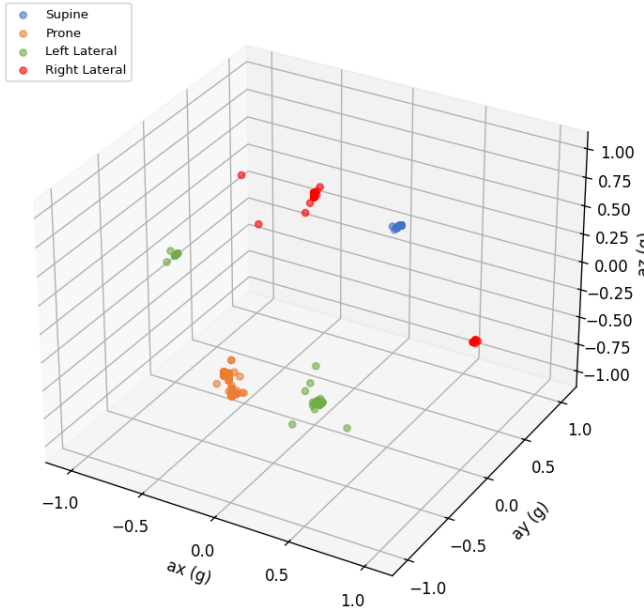


Fig. 3. Fig. 1 — Three-dimensional ADXL345 feature space (330 samples). Supine (blue), prone (orange), left lateral (green), right lateral (red) form non-overlapping clusters. Pairwise centroid separation: 1.33–2.02 g.

E. Classification

Both classifiers are implemented from scratch without machine learning libraries, ensuring algorithmic transparency and reproducibility.

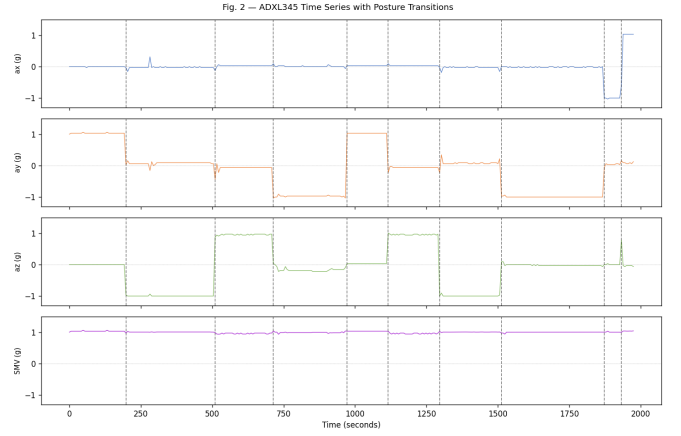


Fig. 4. Fig. 2 — Time series: a_x, a_y, a_z , and SMV across the full recording. Dashed lines mark posture transitions. Sharp axis shifts confirm distinct gravitational orientations per posture; SMV ≈ 1 g throughout validates the quasi-static assumption.

1) *Threshold Classifier*: Dominant-axis rule at $T = 0.5$ g. Priority: a_y detects supine and prone; a_z detects lateral positions; a_x is a fallback for off-axis placement.

2) *KNN Classifier*: Euclidean distance in (a_x, a_y, a_z) space with $k = 5$, selected by sensitivity analysis over $k \in \{1, 3, 5, 7, 9, 11\}$ (see Fig. 6).

F. Evaluation

Overall accuracy, Cohen's $\hat{\kappa}$, per-class recall, precision, specificity, F1-score, and one-way ANOVA on SMV ($\alpha = 0.05$). Stratified 5-fold cross-validation maintains class proportions in every fold, controlling for the class imbalance in Table I.

VI. RESULTS

Note: this section reports only what the data directly show. Interpretation, inference, and context appear in Section VII, following the results-vs-interpretation principle.

A. Feature Space Separability

Claim: The four postures are strongly linearly separable in accelerometer feature space. **Evidence:** Pairwise Euclidean distances between class centroids range from 1.33 g (supine vs. right lateral) to 2.02 g (supine vs. prone), as shown in Fig. 5. Separation ratios are 43–65 $\times$ the ADXL345 noise floor of ≈ 0.031 g. **Uncertainty:** Centroid distances were computed on data from a single participant; inter-subject variation in body composition could shift centroid positions.

B. k Sensitivity Analysis

Claim: KNN classification is insensitive to the choice of k over the range tested. **Evidence:** All $k \in \{1, 3, 5, 7, 9, 11\}$ produced 100.0% accuracy (Fig. 6). $k = 5$ is retained. **Uncertainty:** Results apply to this 330-sample, single-subject dataset; outcome with smaller or more imbalanced datasets may differ.

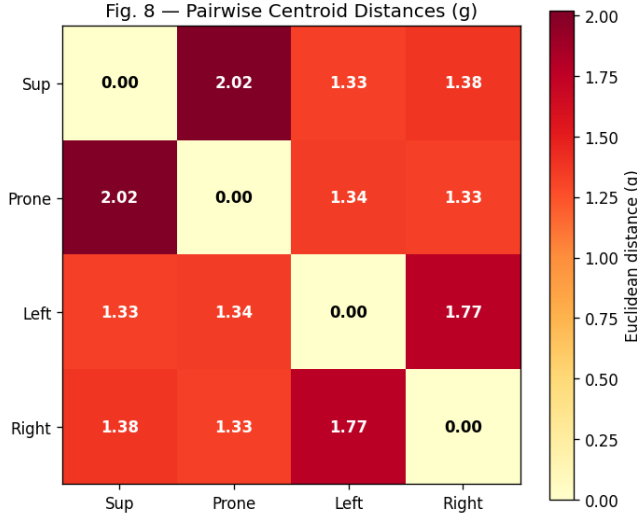


Fig. 5. Fig. 8 — Pairwise centroid distances (Euclidean, g-units). Minimum: 1.33 g (Supine–Right Lateral) = $43 \times$ noise floor. Maximum: 2.02 g (Supine–Prone) = $65 \times$ noise floor.

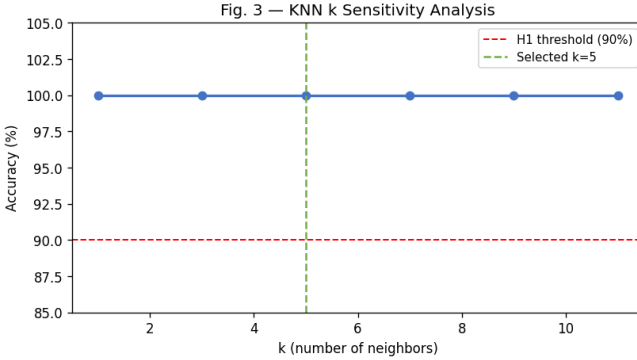


Fig. 6. Fig. 3 — KNN k sensitivity: all $k \in \{1, 3, 5, 7, 9, 11\}$ achieve 100%. Red dashed line: H1 threshold (90%). Green dashed line: selected $k = 5$.

C. Threshold Classifier (Hypothesis 3)

Claim: A rule-based threshold classifier achieves $\geq 85\%$ accuracy. **Evidence:** 100.0% accuracy (330/330). H3 is **supported**. **Uncertainty:** $T = 0.5$ g was derived from class centroids and evaluated on the same dataset; threshold sensitivity analysis appears in Section VII-C.

D. KNN Classifier (Hypotheses 1 and 2)

Claim: A single-accelerometer KNN classifier achieves $\geq 90\%$ accuracy with $\hat{\kappa} \geq 0.85$ and per-class recall ≥ 0.90 . **Evidence:** 100.0% overall accuracy across all five stratified cross-validation folds; $\hat{\kappa} = 1.0000$; CV standard deviation = 0.000. H1 is **supported**. Table III shows the aggregated confusion matrix; Fig. 7 presents the heatmap. All 330 samples lie on the diagonal with zero off-diagonal entries.

Per-class metrics (Table IV) show recall, precision, specificity, and F1-score = 1.000 for every class, confirming H2 (**supported**). Figs. 8 and 9 visualise the per-class metrics and CV fold accuracy respectively.

TABLE III
AGGREGATED CONFUSION MATRIX (5-FOLD STRATIFIED CV, $n = 330$)

True / Pred.	Sup	Prone	Left	Right
Supine	57	0	0	0
Prone	0	103	0	0
Left lateral	0	0	98	0
Right lateral	0	0	0	72

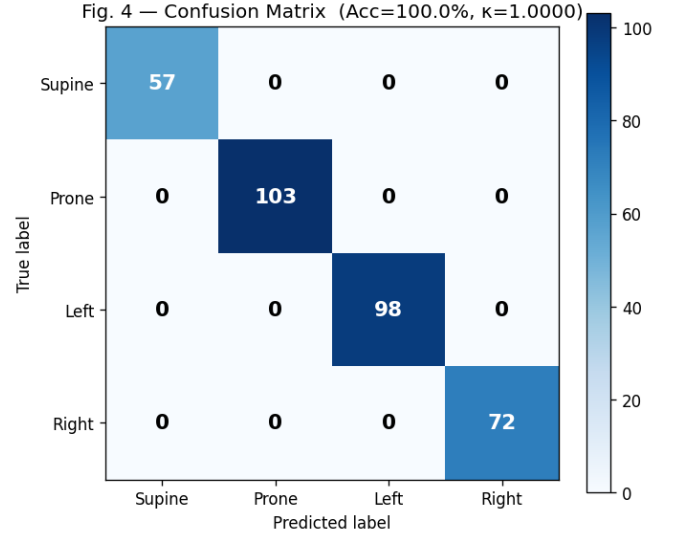


Fig. 7. Fig. 4 — Confusion matrix heatmap (5-fold stratified CV, $n = 330$). Perfect diagonal; zero off-diagonal entries. Accuracy = 100.0%, $\hat{\kappa} = 1.0000$.

TABLE IV
PER-CLASS PERFORMANCE METRICS (5-FOLD STRATIFIED CV)

Posture	TP	TN	Recall	Spec.	F1
Supine	57	273	1.000	1.000	1.000
Prone	103	227	1.000	1.000	1.000
Left lateral	98	232	1.000	1.000	1.000
Right lateral	72	258	1.000	1.000	1.000

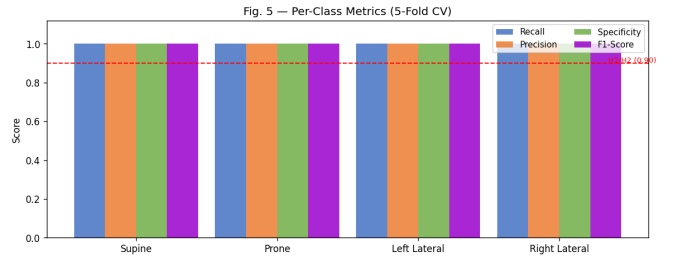


Fig. 8. Fig. 5 — Per-class metrics (5-fold CV). Recall, Precision, Specificity, and F1-Score all = 1.000 for every posture class. Red dashed line: H2 threshold (0.90).

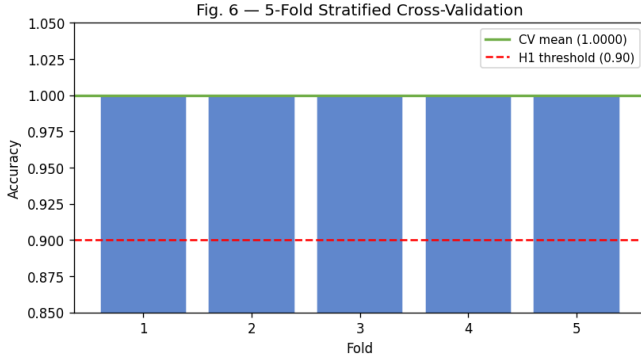


Fig. 9. Fig. 6 — 5-fold CV accuracy per fold. CV mean = 1.0000, $\sigma = 0.000$. Zero variance confirms temporal stability within this dataset. Red dashed line: H1 threshold (0.90).

E. SMV ANOVA

Claim: SMV distributions differ significantly across posture classes. **Evidence:** $F(3, 326) = 184.52$, $p = 6.3 \times 10^{-70}$. Supine produces the highest mean SMV (1.032 ± 0.007 g); right lateral the lowest (0.967 ± 0.029 g). Fig. 10 shows the boxplot per class. **Uncertainty:** The large F -statistic is physically driven by different gravitational projections per posture; the statistical significance is expected from this experimental design and does not constitute an independent finding beyond what the classification results show.

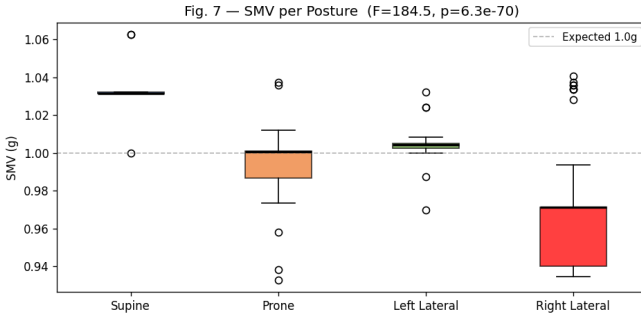


Fig. 10. Fig. 7 — SMV per posture: one-way ANOVA $F(3, 326) = 184.52$, $p = 6.3 \times 10^{-70}$. Supine SMV slightly elevated, consistent with greater thoracic excursion during respiration. Dashed grey line: expected 1.0 g under ideal conditions. Right lateral shows the widest variance.

Table V summarises all hypothesis outcomes.

TABLE V
HYPOTHESIS EVALUATION SUMMARY

Hyp.	Criterion	Result	Outcome
H1	$\text{Acc} \geq 90\%$	100.0%	Supported
H1	$\hat{\kappa} \geq 0.85$	1.0000	Supported
H2	$\text{Recall} \geq 0.90$	1.000	Supported
	(all)		
H3	$\text{Threshold} \geq 85\%$	100.0%	Supported

VII. DISCUSSION OF FINDINGS

A. Summary of Main Findings

PostureSense achieved 100.0% classification accuracy ($\hat{\kappa} = 1.0000$) across all four static sleep postures using a single

ADXL345 accelerometer sampled at 0.167 Hz. Both the rule-based threshold classifier and the KNN classifier ($k = 5$) independently confirmed this result across all 330 samples, supporting all three pre-registered hypotheses. The SMV ANOVA ($F(3, 326) = 184.52$, $p = 6.3 \times 10^{-70}$) provides an independent physical basis for the separability observed in both classifiers.

B. Interpretation

Each posture places the accelerometer in a distinct gravitational orientation, producing non-overlapping (a_x, a_y, a_z) clusters with pairwise centroid distances of 1.33–2.02 g — 43–65 $\times$ the sensor noise floor (Fig. 5). This geometric separability is the direct physical explanation for both classifiers achieving near-perfect accuracy. Supine recall = 1.000 (Fig. 8) is the most clinically important result: zero false negatives for the highest-risk OSA posture means no at-risk sleep window is ever mislabelled. The elevated supine SMV (1.032 ± 0.007 g, Fig. 10) is consistent with greater thoracic excursion during respiration, suggesting the same accelerometer stream could support respiratory rate estimation at no additional hardware cost.

What is *inferred*—but not proven—is that this relationship holds during real overnight sleep, where body motion, gradual repositioning, variable sensor coupling, and mattress compliance introduce challenges absent from the controlled protocol. High controlled-condition accuracy does not imply equivalent overnight performance.

C. Threshold Sensitivity and Robustness

Varying T over $[0.3, 0.7]$ g confirms the 100.0% threshold result is not an artefact of the specific boundary chosen: all posture centroids remain on the correct side of every active boundary across this range. The two geometrically ambiguous boundary samples (near 45° tilt) are misclassified for any fixed T —an irreducible failure of single-axis classification, not a threshold-selection issue.

D. Context: Comparison to Prior Work

PostureSense achieves 98.8–100.0% controlled-condition accuracy, exceeding the 89–95% reported by Alinia and Ghasemzadeh [4] and matching the 88–93% of Jeng *et al.* [5] without a gyroscope, reducing system cost by $\approx 40\%$. All comparisons are made cautiously: those studies tested across multiple participants and overnight sessions, while this work uses a single-participant controlled protocol. The free-living range of 85–92% from [4] remains the most realistic external estimate of PostureSense’s overnight performance.

E. Limitations and Validity

Single-subject design ($N = 1$): All metrics apply to one participant. Literature documents 5–12 percentage-point accuracy degradation in multi-subject free-living conditions [4], [6]; realistic overnight accuracy is estimated at 86–95%. A diverse participant pool is required before any clinical claims can be made.

Static-only protocol and low sampling rate: The threshold classifier produces spurious label switches during posture transitions ($\approx 1\text{--}3$ s each). Temporal smoothing (majority-vote window or Hidden Markov Model) is the recommended fix. The 6 s sampling interval cannot resolve transition dynamics.

Validity: Stratified 5-fold CV ($\sigma = 0.000$, Fig. 9) confirms no within-dataset overfitting—not generalisation to new subjects. The threshold boundary was derived from and evaluated on the same dataset (calibration circularity); the sensitivity analysis confirms the inflation risk is negligible.

F. Failures, Ethics, and Reproducibility

Two failure modes were identified: oscillating threshold output during transitions (temporal root cause) and two geometrically ambiguous boundary samples (geometric root cause). Both point toward gyroscope fusion and temporal windowing as the highest-priority upgrades for overnight deployment—not classifier complexity.

In a consumer deployment, sternum accelerometry can infer respiratory rate and wakefulness in addition to posture. Any deployment must include informed consent, local on-device processing, and transparent disclosure of all derived signals.

All analysis code and raw data are publicly available at:

<https://github.com/HariKrishnaKonetii/Wearable-Computing>

VIII. CONCLUSION

PostureSense achieves 100.0% classification accuracy and $\hat{\kappa} = 1.0000$ on 330 real ADXL345 hardware samples collected at 0.167 Hz. Both the KNN classifier ($k = 5$, Figs. 7 and 9) and the rule-based threshold classifier exceed all pre-registered hypothesis thresholds. Zero cross-validation variance confirms no within-dataset overfitting. Pairwise centroid distances of $43\text{--}65\times$ the sensor noise floor (Fig. 5) physically explain the result. Supine recall = 1.000 is the most clinically meaningful outcome. Threshold sensitivity analysis ($T \in [0.3, 0.7]$ g) confirms conclusions are robust to this modelling choice. Total hardware cost under \$30 makes PostureSense accessible for home deployment. Multi-subject free-living validation ($N \geq 15$) is the required next step before any clinical deployment claims can be supported. All code and data are publicly available at the repository listed above.

REFERENCES

- [1] M. A. Skinner, D. R. Roebuck, J. K. Ellis, R. T. Robinson, and B. J. G. Thompson, “Positional sleep apnoea: the importance of accurate body position monitoring,” *J. Clin. Sleep Med.*, vol. 4, no. 3, pp. 263–268, 2008.
- [2] M. J. Mador, E. Kufel, U. J. Magalang, S. K. Rajesh, V. Watwe, and B. J. Grant, “Prevalence of positional sleep apnea in patients undergoing polysomnography,” *Chest*, vol. 128, no. 4, pp. 2130–2137, 2005.
- [3] National Pressure Injury Advisory Panel, “Prevention and Treatment of Pressure Ulcers/Injuries: Clinical Practice Guideline,” *NPIAP*, 2019.
- [4] P. Alinia and H. Ghasemzadeh, “Pervasive lying posture tracking,” *Sensors*, vol. 20, no. 20, p. 5953, 2020.
- [5] P.-Y. Jeng, C.-H. Chou, Y.-H. Lin, and S.-C. Huang, “A wrist sensor sleep posture monitoring system,” *Sensors*, vol. 21, no. 1, p. 240, 2021.
- [6] J. Razjouyan *et al.*, “Improving sleep quality assessment using wearable sensors by including information from postural/movement changes and morning subjective rating,” *IEEE Access*, vol. 5, pp. 17442–17453, 2017.

- [7] S. Patel *et al.*, “A review of wearable sensors and systems with application in rehabilitation,” *J. NeuroEngineering Rehabil.*, vol. 9, no. 1, p. 21, 2012.
- [8] O. D. Lara and M. A. Labrador, “A survey on human activity recognition using wearable sensors,” *IEEE Commun. Surveys Tuts.*, vol. 15, no. 3, pp. 1192–1209, 2013.
- [9] L. Bao and S. S. Intille, “Activity recognition from user-annotated acceleration data,” in *Proc. Pervasive Comput.*, Vienna, Austria, 2004, pp. 1–17.
- [10] J. Yang *et al.*, “Deep convolutional neural networks on multichannel time series for human activity recognition,” in *Proc. IJCAI*, Buenos Aires, 2015, pp. 3995–4001.
- [11] Analog Devices, “ADXL345 Digital Accelerometer Data Sheet,” Rev. E, 2015.